

Zeev — Tool & Command Cheat Sheet

Everything you drive Zeev with: launch modes, terminal slash commands, device-mode voice intents (in the order they are matched), LLM tools, and the backends behind them. Generated 2026-07-31 from zeev/zeev.py — not from the docs.

1 • Launch modes

python3 zeev/zeev.py	Terminal REPL. All slash commands below live here.
python3 zeev/zeev.py --web	Web UI + JSON/SSE API on HTTP :5000.
python3 zeev/zeev.py --https	Same, HTTPS :5443.
python3 zeev/zeev.py --device	Whisplay HAT device mode — wake word, face, cameras, BT, reminders. Needs the HAT.
python3 zeev/zeev.py --call <NUMBER> [--intent "TEXT"]	Place an HFP phone call over a paired handset and run the call loop.
... --no-greeting	Suppress the spoken startup greeting (works with any mode).
python3 zeev/quantum_convo.py --name NAME [--about T] [--call N]	Quantum-weighted conversation topics, optionally dialled out.
python3 zeev/import_sefaria.py	Populate data/torah.db (resume-safe, ~75 min for the full corpus).
python3 zeev/weekly_reflection.py	Synthesize the last 7 days into a reflection (cron: Sun 07:00).
python3 zeev/quantum_daily.py	One human-dilemma scenario per day (cron: 06:00).
python3 zeev/gcal_auth.py	Interactive Google consent. Re-run after any calendar-scope change or writes 403.

2 • Terminal slash commands

/clear	Wipe the in-memory session	/vol /vol N /vol + /vol -	Read or set speaker volume (%)
/forget	Drop all stored facts	/tts	Toggle spoken replies
/model	Show / pick the model (1 · 8B, 2 · 70B, 3 · GPT-OSS 120B)	/stop	Stop music playback
/memory	List what Zeev knows about you	/joke	Adult joke; also fires on natural phrasing
/memorize	Extract facts from this session now	/look	Camera snapshot → vision model
/forget-fact N	Remove fact N	/flip	Flip the camera image
/note <text>	Save a note	/listen	Record mic → Whisper → treat as input
/notes	List notes	/thermal	MLX90640 thermal frame
/forget-note N	Delete note N	/gps	Wi-Fi/IP location fix with accuracy + ISP
/lang en he es ru auto	Force reply language. auto = English — there is no character-set auto-detect any more	/quantum <idea>	Idea → quantum circuit → interpretation
		/bt ...	bare /bt lists paired · scan · pair N · N connect · off · tether [N] · tether off
		/voice /voice-coach	Voice-coaching mode
		quit exit bye	Extract memory, say goodbye, exit

Joke triggers are multilingual. English (“tell me a joke”, “make me laugh”), Spanish (cuéntame un chiste), Russian (расскажи анекдот), Hebrew (ספר לי בדיחה) and Chinese (讲个笑话) all match _JOKE_RE.

3 · Device-mode voice commands — in match order

`handle_transcript()` walks these gates top-to-bottom and **the first match wins**. If an utterance did something unexpected, read down this list: something above swallowed it.

#	Gate	Say something like	What it does
1	Voice coach	“voice training”, “coach my voice”, “help with my diction”	Enters/continues coaching mode.
2	Language switch	“speak Russian”, “answer in Spanish”, “switch to English”	Sets FORCED_LANG and confirms aloud. Verb must be within 20 chars of the language name.
3	Goodnight	“goodnight”, “sweet dreams”, “I'm going to bed”	The one branch answering in two voices: Zeev (daniel) then Sarina (af_heart). Must be in the first 40 chars.
4	Jokes	“tell me a joke”, “make me laugh”, “say something funny”	Local joke DB, language-matched. Anchored to the start of the utterance.
5	Detail expansion	“tell me more”, “go on”	Speaks the prefetched long form of the previous answer.
6	Bluetooth	“scan for headphones”, “pair the earbuds”, “connect bluetooth”, “disconnect the headphones”, “switch to the speaker”	Scan/pair/connect/disconnect. Scan takes ~10s.
7	BT tethering	“use my phone's internet”, “start tethering”, “stop tethering”	Brings the phone's data link up or down over BT-PAN.
8	Phone call	“dial 617-555-0134”, “ring my wife”, “phone mom”	The trigger is “dial”, not “call” , and must appear in the first 60 chars. Detects voicemail / IVR / live person and adapts.
9	Volume	“turn it up”, “louder”, “lower the volume”	±10% steps on the active output.
10	Named subject	“check on Smokey”, “where's Smokey”	Sweeps the RTSP cameras looking for that subject. Answers “I didn't see ...”, never “he isn't there”. Names come from ZEEV_SUBJECTS in the Pi's .env.
11	House cameras	“check the bedroom cam”, “what's going on upstairs”, “check all cameras”	One Wyze RTSP frame → vision model, ~29s, so it says “Let me look at the ...” first. A sweep is capped at 3 cameras. An ambiguous room gets “Which camera?” rather than a guess.
12	Pi camera	“what do you see”, “take a picture”, “look around”	Only when a Pi camera is attached; otherwise falls to the “Which camera?” ask.
13	Visual effects	“show me fire”, “run the matrix effect”, “psychedelic mode”	12s on the LCD: fire · matrix · psychedelic · liquid · tunnel · plasma · cartoon · starfield · bounce.
14	Quantum	“quantum: should I take the job”, “run a quantum circuit on ...”	Idea → circuit → interference outcomes → interpretation. Must start the utterance.
15	Music	“play Miles Davis”, “put on some jazz” / “stop the music”, “pause it”	Says “Looking for X”, then resolves in the background (~45s for yt-dlp). Must start the utterance.
16	Tools	“remind me to ...”, “set a timer for 10 minutes”, “what are my reminders”, “put dinner on my calendar Thursday at 7”, “make a note”	Pre-gated by _T00L_INTENT_RE and routed to 70B — the 8B is unreliable at tool calls. Slower: this round trip cannot stream.
17	Chat	everything else	Model auto-routed, streamed sentence-by-sentence to TTS. Search / weather / Torah / calendar / GPS context is added to the prompt automatically.

4 · Trigger rules worth memorising

Wake word	The local openWakeWord path listens for “Hey Ze'ev” (daniel voice) and “Hey Sarina” (af_heart voice) — models named in <code>OWW_MODEL_PATH/OWW_VOICE_MAP</code> in the Pi's <code>.env</code> , which is not in git. The <i>cloud fallback</i> path (used only when no model loads) still substring-matches “miss minutes”.
Speak, pause, then ask	On the local path the wake phrase fires on its own and the follow-up is recorded separately. Only the cloud fallback can catch “<wake word>, what's the weather” in one breath.
“dial”, never “call”	“Call” is far too common in speech, so it was dropped. The trigger must sit in the first 60 characters.
Anchored gates	Jokes, music and quantum must <i>start</i> the utterance. Goodnight must be in the first 40 chars; a call trigger in the first 60; a subject name within 40 chars of its trigger.
“Say it in Hebrew” ≠ switch	An object between the verb and the language name (“it”, “the prayer”) makes it a <i>content</i> request — Zeev answers in Hebrew for that turn only. “Say it in English” is deliberately exempt: <code>FORCED_LANG</code> is sticky, and over-triggering toward English is the safe direction.
Reminder beats camera	“Remind me to check on Smokey at four” is a reminder — the subject and camera gates explicitly reject tool phrasing.
Naming a room beats “all”	“Check all the cameras in the basement” is a question about the basement, not a sweep.
Interrupt	The HAT button (GPIO17) cancels speech mid-sentence, and also interrupts while Zeev is still <i>thinking</i> .

5 · LLM tool calls (Zeev's only write capabilities)

Tool	Arguments	Notes
<code>set_reminder</code>	text, when	Schedule a spoken reminder/timer. when is ISO-8601 or “in 10 minutes”; a bare time already past rolls to tomorrow. Polled every 20s; never speaks over a live turn.
<code>list_reminders</code>	–	Pending reminders.
<code>cancel_reminder</code>	id	Cancel by id from <code>list_reminders</code> .
<code>create_calendar_event</code>	summary, when, duration_minutes	Real Google Calendar write. Needs the <code>calendar.events</code> scope — a stale read-only token keeps reads working while writes 403.
<code>add_note</code>	text	Durable note.

6 · Model routing (automatic, no extra API call)

Key	Model id	When it is picked
1	llama-3.1-8b-instant	Fast — default for casual chat. 600 max tokens.
2	llama-3.3-70b-versatile	Smart — code, writing, explanation, and every tool call. Triggered by write/explain/compare/how-to vocabulary. 1200 tokens.
3	openai/gpt-oss-120b	Reasoning — maths, logic, puzzles, spoken arithmetic, percentages, “how many ... 25”, elapsed-time questions. 1200 tokens.

On a 429: a per-model 5-minute cooldown keeps an 8B call working while 70B is limited, then OpenRouter's free tier, then Ollama on bosgame. Device mode also retries 70B/GPT-OSS failures on 8B before showing an error.

7 • What is actually doing the work

Job	Backend	What to know
Speech → text	Groq Whisper large-v3-turbo	Cloud. Call audio gets a vocabulary-biased prompt because 8 kHz SCO makes Whisper hallucinate.
Wake word	openWakeWord 0.4.0 (onnx)	On-device, ~70% of one core, 166 MB. Energy gate skips quiet frames; 0WW_ENERGY_MULT=2.5 on the Pi.
Speech gate	webrtcvad (aggressiveness 3)	Rejects tones and hiss before paying for Whisper. Needs webrtcvad-wheels on Python 3.13.
TTS — English	Kokoro on bosgame + feiergente01	Sarina = af_heart, Zeev = am_adam. Sentences alternate between the two backends. Groq Orpheus daniel is the brain voice.
TTS — ru / es	Piper on bosgame	irina (ru) / daniela (es). Local Piper on the Pi is ~2.6× real time, so it is only the fallback.
TTS — he	gTTS + mpg123	Torah/Sefaria replies force Hebrew TTS regardless of the reply's script.
TTS — last resort	Piper local → espeak-ng	Used only when everything above fails.
Vision	OpenRouter free tier	Gemma 4 26B first, then 31B, then Nemotron. ~21-25s per image — this, not the frame grab, is the whole latency.
Web search	Tavily	Behind needs_search; weather is a strict subset, and units get spelled out for speech.
Chat fallback	Ollama on bosgame	llama3.1:8b for chat, llama3.2:1b for memory extraction, nomic-embed-text for semantic recall.
Memory	SQLite data/zeev.db (WAL)	messages · facts · notes · reminders · settings · message_vecs · reflections · quantum_insights.
Torah RAG	SQLite FTS5 data/torah.db	Tanakh, Mishna, Talmud, Apocrypha, Siddur, Zohar, Dead Sea Scrolls, Sumerian. Chapter granularity only — a parsha starting mid-chapter has its real bounds stated, not trimmed.
Music	yt-dlp → ffmpeg → mpg123	Must be the upstream binary in /usr/local/bin; the apt build can no longer extract audio. Auto-updated Sundays 04:00.
Cameras	Wyze RTSP firmware	Only two cameras work: smokeys-cam (10.0.0.217) and bedroom-cam (10.0.0.84), both on /stream0. The other six are blocked on a TUTK/DTLS handshake and cannot be flashed.
Location	Wi-Fi APs → Google → beaondb → IP	±11 m with a Wi-Fi fix. City is only named under 5 km accuracy; above that, region only.
Audio daemon	zeev-audio (Go, arm64)	NDJSON over /tmp/zeev-audio.sock. Owns TTS, recording, BT, music, SCO. Python falls back to its own subprocesses if the socket is down.

8 • Ops one-liners

./deploy.sh	Deploy to the Pi. Commit first — it relies on git push origin main.
cd zeev-audio && make pi	Cross-compile the Go daemon for arm64.
scp bin/zeev-audio-pi ragnar@ragnarok:~/zeev-audio/zeev-audio	Stop the service first, copy, then start it.
ssh ragnar@ragnarok	The Pi. LAN 10.0.0.168, or ragnarok.tail9c2c7c.ts.net over Tailscale.
journalctl -u zeev-device -f	Live device log. -u zeev-audio for the daemon; -b -1 reaches the previous boot.
systemctl status zeev-device zeev-audio	Read the actual state before assuming it.
yt-dlp --version	First thing to check when music silently stops working.
pytest	tests/ pins the intent gates, model routing, stage-direction stripping, RGB565 conversion and the camera guards.

Cheat sheet generated for Alex · sources: zeev/zeev.py, zeev-audio/, CLAUDE.md · trigger phrasing is regex-matched, so the examples are representative, not exhaustive.